DharaScope: Satellite Surface Water Extent Report

Sentinel-2 MSI Level-2A Feasibility Assessment · Gomti River, Lucknow, UP · Execution: 2026-10-28T04:21:13Z

PROVENANCE STATUS: VALIDATED REAL SATELLITE DATA
Provider: PUBLIC_OPEN_SENTINEL2_COGS — Public Open Sentinel-2 Level-2A Cloud-Optimized GeoTIFF Archive (AWS Registry of Open Data / ESA)

Study Reach: Gomti River, Lucknow	**CRS:** EPSG:32644	**Resolution:** 10m (100.0 m²/px)
Study Bounding Box: [80.88, 26.8, 81.0, 26.9]	**NDWI Threshold:** $\tau = 0.00$	**Evaluated Area:** 13,230.37 ha
Observation T1: 2023-10-10 (0.0% cloud)	**Observation T2:** 2023-12-09 (0.0% cloud)	**Interval:** 60 calendar days

1. Quantitative Surface Water Dynamic Metrics

Hydrological Classification Category	Pixel Count	Area (m²)	Area (ha)	% of T1 Baseline
Earlier Inundation (T1)	18,456	1,845,600	184.56	100.00%
Later Inundation (T2)	13,933	1,393,300	139.33	75.49%
Perennial Channel Core (Persistent)	12,518	1,251,800	125.18	67.83%
Seasonal Channel Recession (Apparent Loss)	5,938	593,800	59.38	32.17%
Localized Inundation Expansion (Apparent Gain)	1,415	141,500	14.15	7.67%
Net Surface Extent Delta (T2 - T1)	-4,523	-452,300	-45.23	-24.51%

2. Multi-Temporal Cartographic Visualization

Figure 1: Synoptic 4-panel analysis illustrating continuous NDWI distributions, binary water masks, and the spatial transition matrix over Lucknow.

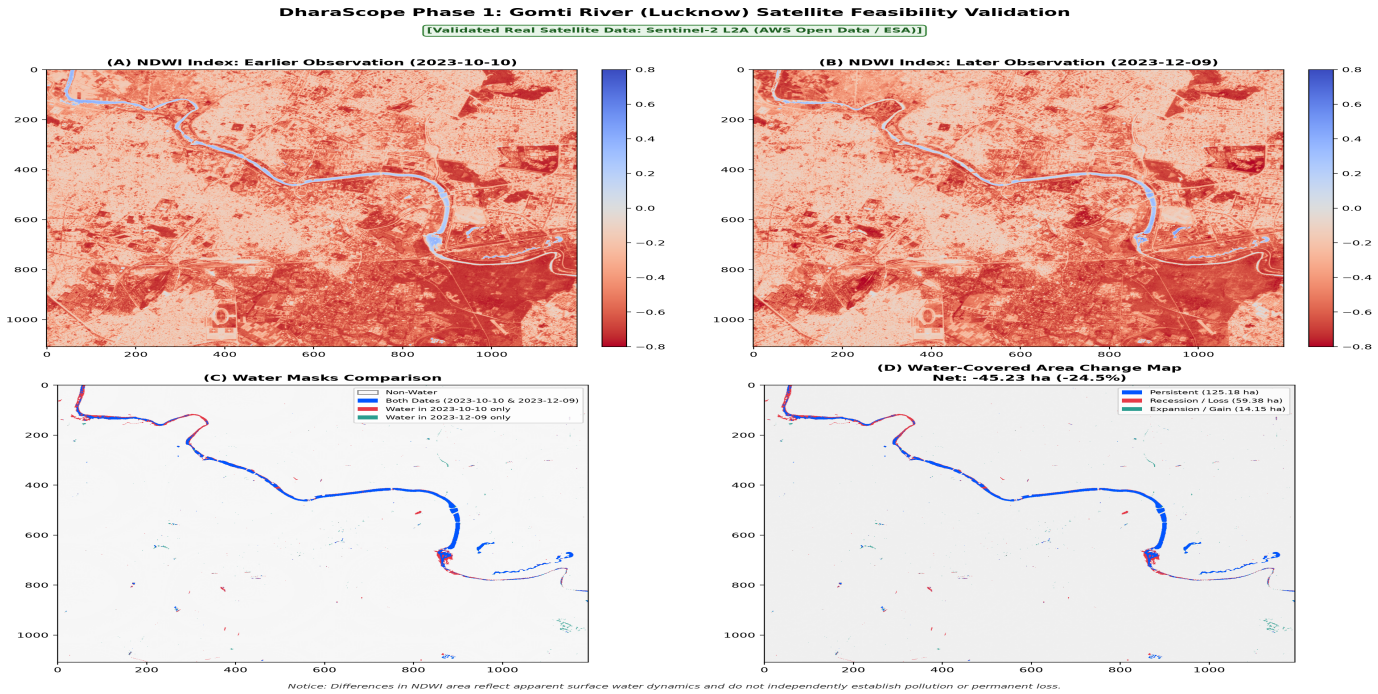
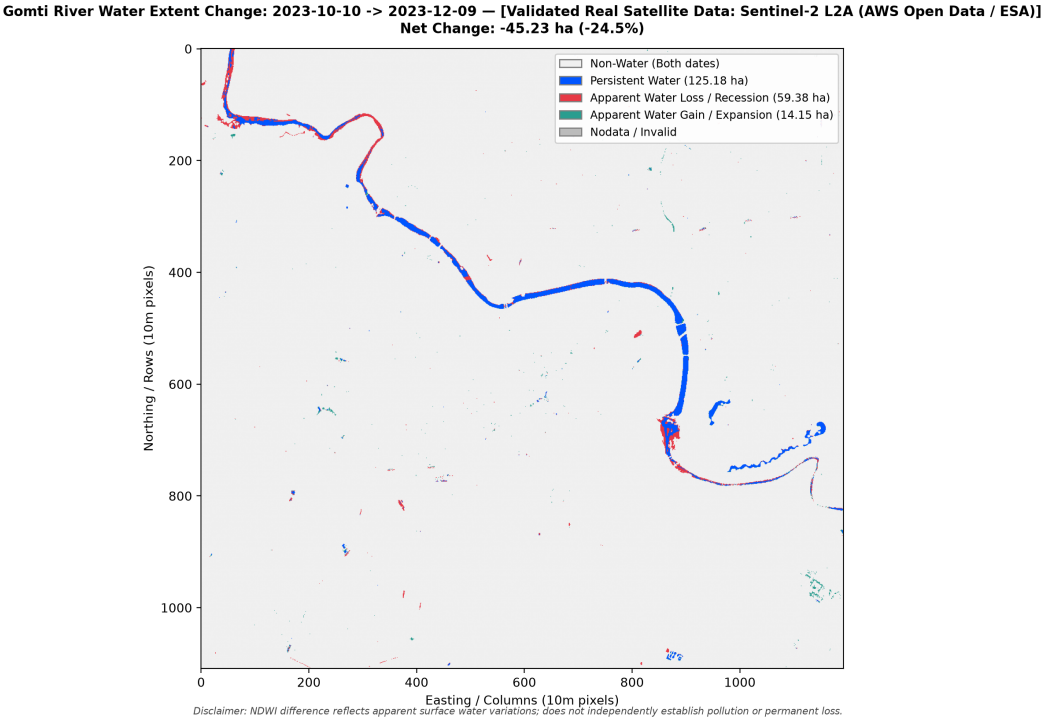


Figure 2: Four-class categorical water transition map.



3. Spectral Index Formulation & Analytical Methodology

Normalized Difference Water Index (NDWI) [McFeeters, 1996]:
Surface water boundaries are delineated using the normalized difference of green and near-infrared reflectances: NDWI = (Band 3 [Green] - Band 8 [NIR]) / (Band 3 [Green] + Band 8 [NIR]).
Liquid water exhibits high reflectance in optical green wavelengths (560 nm) and near-total absorption in near-infrared (842 nm), whereas terrestrial vegetation and urban soils strongly reflect NIR radiation. Pixels satisfying NDWI > 0.00 across common valid pixels are classified as open water.
Numerical Protections: Denominators satisfying |B03 + B08| ≤ 1e-6 are masked as invalid to eliminate division by zero, and valid ratios are constrained to the theoretical physical domain [-1.0, 1.0].

4. Hydrological Limitations & Evaluation Disclaimer

ACADEMIC NOTICE: *Scientific Limitation Notice: Differences in NDWI-classified water-covered area reflect apparent surface water extent fluctuations. These fluctuations are primarily governed by seasonal precipitation, upstream dam releases, agricultural withdrawal, channel morphology, and riparian vegetation growth. NDWI-based area changes do NOT independently establish permanent water loss, environmental degradation, or river pollution without calibrated in-situ hydrological gauging, bathymetric surveys, and water quality sampling.*

Hydrological Context: *The Gomti River basin is monsoonal with peak precipitation during July–September. The observed 24.51% surface water recession between October and December corresponds to post-monsoon baseflow decline, upstream Gaughat Barrage stage modulation, and the emergence of riparian sandbars and seasonal silt banks. Surface area differences do NOT indicate river desiccation or water contamination.*